

PRACTICAL 4:

(a)

```
harshika@siri: ~  
for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc practical4.c -o practical2  
harshika@siri:~$ ./practical4  
-bash: ./practical4: No such file or directory  
harshika@siri:~$ gedit practical4.c  
  
** (gedit:2691): WARNING **: 04:15:21.880: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:2691): WARNING **: 04:15:21.880: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc practical4.c -o practical4  
harshika@siri:~$ ./practical4  
Child 1: PID = 2882  
Child 2: PID = 2883  
Parent waiting using wait()  
Child 3: PID = 2884  
Child 1 completed  
A child process terminated  
Child 2 completed  
A child process terminated  
Child 3 completed  
A child process terminated  
All child processes completed  
harshika@siri:~$ gedit practical4.c  
  
** (gedit:2906): WARNING **: 04:17:09.785: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:2906): WARNING **: 04:17:09.786: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
1#include <stdio.h>  
2#include <stdlib.h>  
3#include <unistd.h>  
4#include <sys/wait.h>  
5int main() {  
6    int i;  
7    pid_t pid;  
8    for (i = 0; i < 3; i++) {  
9        pid = fork();  
10       if (pid == 0) {  
11           printf("Child %d: PID = %d\n", i + 1, getpid());  
12           sleep(2 + i);  
13           printf("Child %d completed\n", i + 1);  
14           exit(0);  
15       }  
16   }  
17   printf("Parent waiting using wait()\n");  
18   for (i = 0; i < 3; i++) {  
19       wait(NULL);  
20       printf("A child process terminated\n");  
21   }  
22   printf("All child processes completed\n");  
23   return 0;  
24 }
```

(b)

```
harshika@siri: ~  
  
** (gedit:2906): WARNING **: 04:17:09.786: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gedit practical4.1.c  
  
** (gedit:3368): WARNING **: 04:19:42.089: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:3368): WARNING **: 04:19:42.089: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gedit practical4.1.c  
  
** (gedit:3545): WARNING **: 04:20:10.203: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:3545): WARNING **: 04:20:10.204: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc practical4.1.c -o practical4.1  
harshika@siri:~$ ./practical4.1  
First child: 3873  
Second child: 3874  
First child finished  
Second child finished  
harshika@siri:~$ gedit practical4.1.c  
  
** (gedit:3884): WARNING **: 04:20:38.184: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:3884): WARNING **: 04:20:38.184: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
1#include <stdio.h>  
2#include <stdlib.h>  
3#include <unistd.h>  
4#include <sys/wait.h>  
5int main() {  
6    pid_t pid1, pid2;  
7    pid1 = fork();  
8    if (pid1 == 0) {  
9        printf("First child: %d\n", getpid());  
10       sleep(3);  
11       exit(0);  
12   }  
13   pid2 = fork();  
14   if (pid2 == 0) {  
15       printf("Second child: %d\n", getpid());  
16       sleep(5);  
17       exit(0);  
18   }  
19   waitpid(pid1, NULL, 0);  
20   printf("First child finished\n");  
21   waitpid(pid2, NULL, 0);  
22   printf("Second child finished\n");  
23   return 0;  
24 }
```

Zombie Process:

```
harshika@siri: ~  
First child: 3873  
Second child: 3874  
First child finished  
Second child finished  
harshika@siri:~$ gedit practical4.1.c  
  
** (gedit:3884): WARNING **: 04:20:38.184: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:3884): WARNING **: 04:20:38.184: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gedit zombie.c  
  
** (gedit:4350): WARNING **: 04:45:55.683: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:4350): WARNING **: 04:45:55.684: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc zombie.c -o zombie  
harshika@siri:~$ ./zombie  
Parent process (PID: 4841) sleeping...  
Child process (PID: 4842) exiting...  
harshika@siri:~$ gedit zombie.c  
  
** (gedit:4862): WARNING **: 04:47:43.414: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:4862): WARNING **: 04:47:43.414: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
1#include <stdio.h>  
2#include <stdlib.h>  
3#include <unistd.h>  
4  
5int main() {  
6  
7    pid_t pid = fork();  
8  
9    if (pid == 0) {  
10  
11        printf("Child process (PID: %d) exiting...\n", getpid());  
12        exit(0);  
13    }  
14  
15    else {  
16  
17        printf("Parent process (PID: %d) sleeping...\n", getpid());  
18        sleep(30);  
19    }  
20  
21    }  
22  
23    return 0;  
24 }
```